《计算机网络》课程作业（5）

（3月8日布置；3月17日课前手写提交）

1. Assume that two computers are using time division multiplexing to take turns sending 1500-byte packets over a shared channel that operates at 10Mbps.

    a. If the hardware takes 100 microseconds after one computer stops sending before the other can begin, how long will it take for each of the computers to send a one gigabyte data file?

    b. Calculate the time required for transmission if the computers transmit serially. (Assume a minimum delay of 5 microseconds between two packets sent from the same computer.)

2. What are the main differences among broadcast and multicast?

3. 请从分组和交换技术的作用解释局域网使用分组交换技术的必要性。

4. 请作图描述以太网的帧格式，标注每个部分的长度及含义。

5. 通过 Omnipeek 或类似软件观察，访问不同外网站点时数据包的目的 MAC 地址是否有变化？能否以此在宿舍区观察到软件学院网站的物理地址？请简要解释。